

Atharva Prabhu

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Summary

AI Engineer who owns technical design and delivery across **6 repositories**, shipping **tenant-scoped tool-calling LLM assistants**, **hybrid-retrieval RAG services**, and an **Anthropic-compatible model gateway** with automatic fallback routing. Builds evaluation into the development loop, including a **221-question RAG benchmark run in CI**, and implements **guardrails** that keep proprietary information out of public-facing assistant responses. Adds MLOps depth in **MLflow experiment tracking**, **drift-triggered retraining**, and **Pydantic-validated FastAPI services**, with fluency in **Python, SQL and Docker**.

Technical Skills

Languages: Python, SQL, TypeScript, C/C++, R, Bash/Linux

LLMs & Agents: Claude API, Tool Calling, LangChain, OpenAI GPT-4, Google Gemini, Ollama, Prompt Engineering, Guardrails

Retrieval & RAG: Hybrid Retrieval, Cross-Encoder Reranking, RAG Evaluation, FastEmbed, Sentence Transformers, Pinecone, FAISS

MLOps & Observability: MLflow, Evidently AI, Prometheus, GitHub Actions, CI/CD, TensorBoard, Blue/Green Deploys

APIs, Data & Infra: FastAPI, Pydantic, Express, WebSockets, Docker, Git, PostgreSQL, TimescaleDB, MySQL, React

ML/DL: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, Computer Vision, NLP

Professional Experience

AI Engineer — *JAN Resources, Odessa, TX*

Apr 2026 – Present

- Own **technical design and implementation** across **6 repositories** for an oilfield chemical company's AI and software products, with **full latitude on architecture and tooling**
- Developed **tenant-scoped, tool-calling LLM assistant (Claude Haiku 4.5)** answering customer questions about their own sites via **TimescaleDB telemetry** queries over up to **3 tool rounds**, with **streamed responses** and **admin review**
- Built **hybrid-retrieval RAG service (bge-m3 embeddings, MiniLM cross-encoder reranker)** over **117 contextualized chunks** from PDF, DOCX and PPTX sources, with a **CLI** for ingestion, querying and **synthetic Q&A generation**
- Designed **RAG evaluation framework of 221 questions** (20 hand-written, 201 generated) in documentation and customer phrasing, tracking the **gap between them** as the key metric and running the benchmark in **CI**
- Built **Anthropic-compatible gateway** routing requests between a **local Ollama model** and **Claude**, with **direct-API fallback** when the gateway is unavailable, deployed as a **resource-capped container** on a shared GPU server
- Launched **public AI sales assistant** with local **Kokoro TTS** and **guardrails** that block the proprietary dosing method, logging prospect questions to steer **sales messaging** toward frequently asked topics
- Fixed **8 security** and **7 performance** issues, including **cross-user cache leaks** and **N+1 queries**, and resolved a failed **blue/green deploy** by batching a **2.4M-row startup backfill** behind an **advisory lock**
- Engineered **closed-loop H₂S dosing controller** (feedforward + PID trim) with **stale-data rejection**, **safe-baseline fallback** and automatic **demotion to shadow mode**, planning a **shadow-first rollout** with an **ML advisory phase**
- Built **multi-tenant SCADA monitoring platform** (React 19, Express 5, TypeScript) ingesting **~20 Ignition tags per site** at **5 s intervals** with **WebSocket streaming**, **site-scoped access control** and **SMS alarms**
- Built **operations system** in **~60 commits over 4 days** replacing per-customer Excel invoicing (**3–4 hours** of weekly manual work) with an **append-only ledger**, **load reconciliation**, and **OpenInvoice export**

AI/ML Intern — *Filmic Technologies (via CSE 611)*

Aug 2025 – Dec 2025

- Architected **5-stage Snakemake pipeline** orchestrating **13 ML models** for archival film quality analysis under **GPU memory constraints**
- Integrated **Synchformer**, **Whisper ASR** and **Tesseract OCR** with custom **PyTorch** detectors, feeding results into **MySQL** and a **React UI**

Technical Projects

AI-Powered Retrieval-Augmented Generation System — *LangChain, GPT-4, Pinecone, FAISS, FastAPI, Docker*

- Architected full-stack **RAG pipeline** with **LangChain**, **OpenAI GPT-4** and **Pinecone**, enabling **semantic search** across **1,000+ documents** with sub-second query responses
- Combined **sentence-transformer embeddings**, **FAISS search** and custom **prompt engineering** for context-aware answers, served via **FastAPI**, **Streamlit** and **Docker**

Energy Demand Forecasting System — *MLflow, Evidently AI, GitHub Actions, FastAPI, Pydantic, Prometheus*

- Built **end-to-end MLOps platform** with **4 daily GitHub Actions workflows** for ingestion, drift monitoring, retraining and data versioning, reaching **MAE of 4,800 MWh** on 24-hour forecasts with **zero manual intervention**
- Implemented **Evidently AI drift detection** (30% thresholds) triggering **automated retraining** and **champion model selection** through an **MLflow registry** across **100+ tracked runs**
- Deployed **FastAPI** service with **Pydantic validation**, **Prometheus metrics** and health checks, serving predictions at **<50ms latency** with **99%+ uptime** and **\$0 infrastructure cost**

GrantMatcherAI – AI-Powered Grant Discovery Platform — *FastAPI, Next.js, PostgreSQL, FastEmbed, Docker*

- Engineered **semantic search pipeline** with **FastEmbed (bge-small-en-v1.5)** and a **hybrid scoring system** blending cosine similarity with **eligibility boosts** to rank federal grants per organization profile
- Built automated **Grants.gov/SAM.gov** ingestion with **audit logs**, **JWT/OAuth2 auth**, and **Docker** deployment on **Render**

Education

MS Computer Science — University at Buffalo, SUNY — **GPA: 3.83/4.0**

Aug 2024 – Dec 2025

Coursework: Deep Learning (A), Machine Learning (A), Pattern Recognition (A-), Operating Systems (A)

BE Computer Science — Savitribai Phule Pune University

Aug 2019 – July 2023